

SYLLABUS INTRODUCTION TO (STATISTICAL) MACHINE LEARNING SUMMER 2026

INDIAN INSTITUTE OF MANAGEMENT BANGALORE

Course Title: Introduction to (Statistical) Machine Learning

Instructor: Sabyasachi Chatterjee

Institution: Indian Institute of Management Bangalore

Lecture Hours: 2:30-5:30 PM on Thursdays.

Office Hours: By appointment

Course Dates: The first class will be held on June 18, 2026. The last regular class will be held on August 20, 2026. An additional make-up class will be scheduled on the week 17th to 20th August.

Course Website: <https://sabyasachic.github.io/courses/iimb-ml-2026/>

Lecture notes, announcements, assignments, and supplementary materials will be posted on the course website.

Course Description and Goals:

Machine learning is often presented as a collection of algorithms for prediction and data analysis. This course takes a different viewpoint. We will study machine learning through the lens of probabilistic inference. We will view machine learning and statistical methods arising from a probabilistic model, typically parametrized by a function class and then we can do statistical inference given data.

Bayesian statistics will serve as a guiding principle throughout much of the course. We will study how uncertainty is represented, updated, and propagated through probabilistic models. At the same time, we will discuss important frequentist tools such as cross-validation, bootstrap, and prediction error estimation.

The goal is not to memorize a laundry list of algorithms. Rather, the goal is to understand a coherent set of ideas connecting statistics, probability, computation, and modern machine learning.

The course is organized around a few major themes:

- Statistical learning and function estimation.
- Bayesian inference, uncertainty quantification and computation.
- Modern generative AI.

Date: June 2026.

Students will come away seeing modern AI not as a disconnected collection of algorithms, but as a coherent development of ideas from statistics, probability, Bayesian inference and function approximation.

Prerequisites:

Students should be comfortable with undergraduate probability, statistics, linear algebra, and multivariable calculus. Some prior exposure to regression and statistical modeling will be helpful but is not strictly required. Basic programming experience in Python is helpful but not required. The course will include computational demonstrations and examples using Python notebooks. Students are free to use LLM's for any coding exercises in the course.

Learning Objectives:

By the end of the course, students should be able to:

- understand machine learning from a probabilistic perspective;
- formulate regression and classification problems using statistical models;
- understand Bayesian inference in simple and moderately complex models;
- understand the role of regularization in high-dimensional regression;
- apply resampling methods such as cross-validation and the bootstrap;
- understand nonlinear regression and function estimation methods;
- understand the basic principles behind neural networks;
- understand the foundations of Bayesian computation;
- understand the key ideas behind modern generative AI models such as variational autoencoders, diffusion models, and transformers.

Topics:

The following list is tentative and may be adjusted depending on the pace of the course.

I. Bayesian Foundations and Supervised Learning

- Bayesian philosophy and probabilistic modeling
- Linear regression
- Logistic regression
- Generalized linear models
- Frequentist and Bayesian perspectives on regression
- Classification using LDA and QDA

II. Model Assessment and Resampling

- Cross-validation
- Bootstrap methods
- Prediction error estimation
- Model selection

III. Bayesian Inference and Empirical Bayes

- Gaussian sequence models
- Empirical Bayes methods
- James–Stein estimation
- Dirichlet–multinomial models
- Bayesian bootstrap

IV. Regularization and High-Dimensional Models

- Ridge regression
- Bayesian interpretation of ridge regression
- Lasso
- Coordinate descent
- Sparsity and variable selection

V. Nonlinear Function Estimation

- Smoothing splines
- Local polynomial regression
- Gaussian process regression
- Trend filtering
- Isotonic regression

VII. Bayesian Computation

- Metropolis–Hastings
- Gibbs sampling
- Hamiltonian Monte Carlo
- Variational inference
- ELBO optimization
- Connections between EM and variational methods

VIII. Modern Generative AI

- Variational autoencoders
- Diffusion denoising probabilistic models
- Transformers
- Large language models
- Probabilistic foundations of generative AI

Course Materials:

Lecture notes will be provided for every class. The course will also include Python notebooks and computational demonstrations. We will fit models, run algorithms, visualize outputs, and examine examples in class.

References:

There is no required textbook for the course. Lecture notes will be the primary source of material. The following references will be useful.

- (1) Aditya Guntuboyina's lecture notes and course materials for *STAT 238: Bayesian Statistics* at UC Berkeley: <https://stat238.berkeley.edu/spring-2026/>.
- (2) James, Gareth, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical Learning: With Applications in Python* (ISLP), Springer.
- (3) Bishop, Christopher M. and Hugh Bishop, *Deep Learning: Foundations and Concepts*, Springer.

Evaluation:

The tentative grading scheme is:

- Homework Assignments: 30%
- Midterm Examination: 30%
- Final Project: 40%

The final project may be completed in groups of ≤ 3 members. Some project ideas are given at the end of this document.

Homework:

Homework assignments are intended to help students engage with both the mathematical and computational aspects of the course. Assignments may include written exercises, conceptual questions, and programming problems.

Collaboration is allowed, but each student must write up their own solutions and submit their own code. Students should clearly indicate the names of any classmates with whom they collaborated.

Final Project

The final project is intended to give students an opportunity to explore one of the major themes of the course in greater depth. Projects may be theoretical, computational, empirical, or a combination of these. Students are encouraged to formulate their own project ideas, subject to instructor approval.

Students will be required to

- submit a Jupyter notebook that fully reproduces all numerical analyses;
- submit a brief written report describing the methodology, results, and conclusions;
- give a 20–25 minute presentation (either in person or as a recorded presentation).

The topics below are intended as suggestions rather than requirements.

1. Comparative Study of Classification Methods. Classification is one of the central tasks in machine learning. In this project, students will compare several classification methods on one or more real datasets.

Possible methods include logistic regression, k -nearest neighbors, linear and quadratic discriminant analysis, decision trees, random forests, gradient boosting, and neural networks.

Students should compare methods using appropriate performance measures such as classification accuracy, precision, recall, F1 score, ROC curves, area under the ROC curve (AUC), and computational cost. The project should investigate how performance depends on the choice of method, training sample size, and tuning parameters, and seek to explain why certain methods perform better than others in different settings.

2. Variational Autoencoders and Diffusion Models. Variational autoencoders (VAEs) and diffusion models are two of the most important classes of modern generative models.

Students should study the probabilistic foundations of VAEs, including latent variable models and the evidence lower bound (ELBO), and explain the basic ideas underlying diffusion models, including the forward noising process and reverse denoising process.

The computational component should involve implementing a moderate-sized diffusion model on a dataset such as MNIST, Fashion-MNIST, CIFAR-10, or another publicly available dataset. Students should investigate how performance depends on key design choices, including network architecture, diffusion schedule, training parameters, and latent dimension, and evaluate the quality of the generated samples.

3. Building a Moderate-Sized Language Model. Large language models are among the most influential developments in modern artificial intelligence. In this project, students will build and study a moderate-sized transformer-based language model.

Students should explain the main components of transformer architectures, including tokenization, embeddings, positional encodings, self-attention, multi-head attention, and autoregressive prediction. The computational component should involve training a transformer model on a publicly available text corpus.

Students should investigate how performance depends on architectural and training choices, such as the number of layers, embedding dimension, number of attention heads, context length, and training corpus size. Performance may be evaluated using validation loss, perplexity, and the quality of generated text.

Additional Project Topics

Students may also choose a project related to Bayesian computation, uncertainty quantification, or modern probabilistic machine learning. Possible topics include:

(1) Bayesian Trend Filtering

Trend filtering is a powerful method for estimating piecewise constant, piecewise linear, and more general piecewise polynomial signals. In this project, students will implement and study a Bayesian version of trend filtering.

Students should formulate a Bayesian model for noisy observations together with a sparsity-inducing prior on the discrete differences of the underlying signal. Possible priors include Laplace priors, scale-mixture representations of Laplace priors, spike-and-slab priors, and related shrinkage priors.

The project should include implementation of an MCMC algorithm for posterior sampling and an empirical study comparing the Bayesian procedure with classical trend filtering. Particular attention should be paid to uncertainty quantification, prior sensitivity, computational efficiency, and convergence of the sampling algorithm.

- (2) **Advanced Markov Chain Monte Carlo.** Implement and study a modern MCMC method such as Hamiltonian Monte Carlo, the No-U-Turn Sampler (NUTS), Riemannian Manifold HMC, or affine-invariant ensemble samplers, and compare its performance with more classical algorithms.
- (3) **Bayesian Optimization.** Implement a Bayesian optimization procedure based on Gaussian processes and study its application to hyperparameter tuning and other black-box optimization problems.
- (4) **Bayesian Quadrature and Probabilistic Numerics.** Study Bayesian approaches to numerical integration and compare them with classical Monte Carlo methods.
- (5) **Applications of Bayesian Inference.** Develop a Bayesian analysis of a scientific, economic, engineering, or social science dataset. Possible topics include hierarchical models, missing-data problems, spatial models, or other applied Bayesian modeling problems.

Students are also welcome to propose their own project topic, provided that it is related to the themes of the course and receives prior approval from the instructor.